

Hofstadter physics in tight-binding model

Tianyu Qiao, Hao Shi

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1 Peierls substitution

For a generic lattice with $\mathbf{a}_1$ and $\mathbf{a}_2$ as the basis vectors, we use $\mathbf{R}$ to denote the Bravais lattice vectors, and use $\boldsymbol{\tau}_\alpha$ to denote the sublattice positions within the first unit cell. Let $\mathbf{r}_{\mathbf{R}\alpha} = \mathbf{R} + \boldsymbol{\tau}_\alpha$ denote the position of orbitals centers. Suppose we have a spinless two-center tight-binding (TB) model (atomic gauge)

$$H = \sum_{\mathbf{R}\alpha} \sum_{\mathbf{R}'\alpha'} h_{\alpha\alpha'}(\mathbf{r}_{\mathbf{R}\alpha} - \mathbf{r}_{\mathbf{R}'\alpha'}) |\mathbf{r}_{\mathbf{R}\alpha}, \alpha\rangle_0 \langle \mathbf{r}_{\mathbf{R}'\alpha'}, \alpha'|_0. \quad (1)$$

The bare translation operator is (for any lattice vector $\mathbf{a} = n_1\mathbf{a}_1 + n_2\mathbf{a}_2$)

$$T(\mathbf{a}) |\mathbf{r}_{\mathbf{R}\alpha}, \alpha\rangle_0 = |\mathbf{r}_{\mathbf{R}\alpha} - \mathbf{a}, \alpha\rangle_0, \quad \langle \mathbf{r}_{\mathbf{R}\alpha}, \alpha|_0 T^\dagger(\mathbf{a}) = \langle \mathbf{r}_{\mathbf{R}\alpha} - \mathbf{a}, \alpha|_0. \quad (2)$$

The bare TB Hamiltonian satisfies the bare lattice translation invariance $T(\mathbf{a})HT^\dagger(\mathbf{a}) = H$.

The Peierls substitution suggests that, under magnetic field $\mathbf{B} = \nabla \times \mathbf{A} = B_z \mathbf{e}_z = \chi B \mathbf{e}_z$ where $B = |\mathbf{B}|$ and $\chi = B_z/B = \pm 1$, the model is modified into

$$H_A = \sum_{\mathbf{R}\alpha} \sum_{\mathbf{R}'\alpha'} h_{\alpha\alpha'}^A(\mathbf{r}_{\mathbf{R}\alpha}, \mathbf{r}_{\mathbf{R}'\alpha'}) |\mathbf{r}_{\mathbf{R}\alpha}, \alpha\rangle \langle \mathbf{r}_{\mathbf{R}'\alpha'}, \alpha'|, \quad (3)$$

$$h_{\alpha\alpha'}^A(\mathbf{r}_{\mathbf{R}\alpha}, \mathbf{r}_{\mathbf{R}'\alpha'}) = h_{\alpha\alpha'}(\mathbf{r}_{\mathbf{R}\alpha} - \mathbf{r}_{\mathbf{R}'\alpha'}) e^{i\theta(\mathbf{r}_{\mathbf{R}\alpha}, \mathbf{r}_{\mathbf{R}'\alpha'})}, \quad (4)$$

where the phase is a line-integral of $\mathbf{A}$ between the two sites

$$\theta(\mathbf{r}_{\mathbf{R}\alpha}, \mathbf{r}_{\mathbf{R}'\alpha'}) = -\frac{e}{\hbar} \int_{\mathbf{r}_{\mathbf{R}'\alpha'}}^{\mathbf{r}_{\mathbf{R}\alpha}} \mathbf{A}(\mathbf{r}') \cdot d\mathbf{r}'. \quad (5)$$

The relation between the new basis $|\mathbf{r}_{\mathbf{R}\alpha}, \alpha\rangle$ and original basis $|\mathbf{r}_{\mathbf{R}\alpha}, \alpha\rangle_0$ will become clear below.

1.1 A derivation of Peierls phase from continuum model

The Peierls substitution can be derived using Wannier functions. Assume a small enough B field, so that $\mathbf{A}$ changes gradually in the lattice ($l_B \gg a$). When $B = 0$, the Hamiltonian $H = \frac{\mathbf{p}^2}{2m} + V(\mathbf{r})$ has Wannier orbital $\phi_\alpha(\mathbf{r} - \mathbf{r}_{\mathbf{R}\alpha}) = \langle \mathbf{r} | \mathbf{r}_{\mathbf{R}\alpha}, \alpha \rangle_0 = \frac{1}{\sqrt{N}} \sum_{\mathbf{k}n} e^{-i\mathbf{k} \cdot \mathbf{r}_{\mathbf{R}\alpha}} U_{n,\alpha}(\mathbf{k}) \psi_{n\mathbf{k}}(\mathbf{r})$, and the hopping reads

$$h_{\alpha\beta}(\mathbf{r}_{\mathbf{R}\alpha} - \mathbf{r}_{\mathbf{R}'\alpha'}) = \int d\mathbf{r} \phi_\alpha^*(\mathbf{r} - \mathbf{r}_{\mathbf{R}\alpha}) [\mathbf{p}^2/(2m) + V(\mathbf{r})] \phi_\beta(\mathbf{r} - \mathbf{r}_{\mathbf{R}'\alpha'}). \quad (6)$$

All $\phi_\alpha(\mathbf{r} - \mathbf{r}_{R\alpha})$ span the complete Hilbert space, but they are inconvenient to use when $B \neq 0$. We instead adopt the gauged orbitals

$$\tilde{\phi}_\alpha(\mathbf{r} - \mathbf{r}_{R\alpha}) = \langle \mathbf{r} | \mathbf{r}_{R\alpha}, \alpha \rangle = \exp\left(-i\frac{e}{\hbar} \int_{\mathbf{r}_{R\alpha}}^{\mathbf{r}} \mathbf{A}(\mathbf{r}') \cdot d\mathbf{r}'\right) \phi_\alpha(\mathbf{r} - \mathbf{r}_{R\alpha}). \quad (7)$$

In such basis we have

$$\begin{aligned} \pi \tilde{\phi}_\alpha(\mathbf{r} - \mathbf{r}_{R\alpha}) &= \exp\left(-i\frac{e}{\hbar} \int_{\mathbf{r}_{R\alpha}}^{\mathbf{r}} \mathbf{A}(\mathbf{r}') \cdot d\mathbf{r}'\right) \left(\pi - e\nabla \int_{\mathbf{r}_{R\alpha}}^{\mathbf{r}} \mathbf{A}(\mathbf{r}') \cdot d\mathbf{r}'\right) \phi_\alpha(\mathbf{r} - \mathbf{r}_{R\alpha}) \\ &\approx \exp\left(-i\frac{e}{\hbar} \int_{\mathbf{r}_{R\alpha}}^{\mathbf{r}} \mathbf{A}(\mathbf{r}') \cdot d\mathbf{r}'\right) \mathbf{p} \phi_\alpha(\mathbf{r} - \mathbf{r}_{R\alpha}), \end{aligned} \quad (8)$$

where the second step is valid when $\mathbf{A}$ changes gradually over the lattice constant and ϕ_α is well-localized. The error of this step is $\sim |\nabla \times \mathbf{A}|a^2 = Ba^2$, which is much smaller when the flux quanta $Ba^2 \ll h/e \Rightarrow a \ll l_B$. Written in these “magnetic Wannier orbitals”, the hoppings are

$$\begin{aligned} h_{\alpha\beta}^A(\mathbf{r}_{R\alpha}, \mathbf{r}_{R'\alpha'}) &= \langle \tilde{\phi}_\alpha(\mathbf{r} - \mathbf{r}_{R\alpha}) | H^A | \tilde{\phi}_\beta(\mathbf{r} - \mathbf{r}_{R'\alpha'}) \rangle \\ &\approx \int d\mathbf{r} \phi_\alpha^*(\mathbf{r} - \mathbf{r}_{R\alpha}) e^{i\frac{e}{\hbar} \int_{\mathbf{r}_{R\alpha}}^{\mathbf{r}} \mathbf{A}(\mathbf{r}') \cdot d\mathbf{r}' - i\frac{e}{\hbar} \int_{\mathbf{r}_{R'\alpha'}}^{\mathbf{r}} \mathbf{A}(\mathbf{r}') \cdot d\mathbf{r}'} \left[\frac{\mathbf{p}^2}{2m} + V(\mathbf{r}) \right] \phi_\beta(\mathbf{r} - \mathbf{r}_{R'\alpha'}) \\ &= e^{-i\frac{e}{\hbar} \int_{\mathbf{r}_{R'\alpha'}}^{\mathbf{r}_{R\alpha}} \mathbf{A}(\mathbf{r}') \cdot d\mathbf{r}'} \int d\mathbf{r} e^{-i\frac{e}{\hbar} \Theta_{\mathbf{r}_{R\alpha}\mathbf{r}_{R'\alpha'}}(\mathbf{r})} \phi_\alpha^*(\mathbf{r} - \mathbf{r}_{R\alpha}) \left[\frac{\mathbf{p}^2}{2m} + V(\mathbf{r}) \right] \phi_\beta(\mathbf{r} - \mathbf{r}_{R'\alpha'}), \end{aligned}$$

where the phase is a triangular-shaped closed loop integral (flux)

$$\Theta_{\mathbf{r}_{R\alpha}\mathbf{r}_{R'\alpha'}}(\mathbf{r}) = \int_{\mathbf{r}_{R'\alpha'} \rightarrow \mathbf{r} \rightarrow \mathbf{r}_{R\alpha} \rightarrow \mathbf{r}_{R'\alpha'}} \mathbf{A}(\mathbf{r}') \cdot d\mathbf{r}'. \quad (9)$$

Since the spread of $\tilde{\phi}_\alpha$ is much smaller than l_B , the main part in (9) should come from the integral of $\mathbf{r}$ around $\mathbf{r}_{R\alpha}$ and $\mathbf{r}_{R'\alpha'}$, i.e., from the region $|\mathbf{r} - \mathbf{r}_{R\alpha}|, |\mathbf{r} - \mathbf{r}_{R'\alpha'}| \sim a \ll l_B$. This makes the triangle very thin and the flux integral tiny. Neglecting $\Theta_{\mathbf{r}_{R\alpha}\mathbf{r}_{R'\alpha'}}(\mathbf{r}) \approx 0$ gives the Peierls substitution.

1.2 Landau gauge and magnetic translation operator

Here we choose the Landau gauge ($\chi = \pm 1$ denotes the sign of magnetic field)

$$\mathbf{A} = (0, \chi Bx, 0), \quad \nabla \times \mathbf{A} = \mathbf{B} = \chi B \mathbf{e}_z. \quad (10)$$

Then the Peierls phase Eq. (5) can be calculated as

$$\theta^\chi(\mathbf{r}_{R\alpha}, \mathbf{r}_{R'\alpha'}) = -\chi \frac{eB}{\hbar} \int_{\mathbf{r}_{R'\alpha'}}^{\mathbf{r}_{R\alpha}} x dy = -\chi \frac{(\mathbf{r}_{R\alpha} + \mathbf{r}_{R'\alpha'})_x (\mathbf{r}_{R\alpha} - \mathbf{r}_{R'\alpha'})_y}{2l_B^2}. \quad (11)$$

The magnetic TB Hamiltonian reads

$$H_A = \sum_{R\alpha} \sum_{R'\alpha'} e^{-i\chi \frac{(\mathbf{r}_{R\alpha} + \mathbf{r}_{R'\alpha'})_x (\mathbf{r}_{R\alpha} - \mathbf{r}_{R'\alpha'})_y}{2l_B^2}} h_{\alpha\alpha'}(\mathbf{r}_{R\alpha} - \mathbf{r}_{R'\alpha'}) |\mathbf{r}_{R\alpha}, \alpha\rangle \langle \mathbf{r}_{R'\alpha'}, \alpha'|. \quad (12)$$

We now find the magnetic translation operator. To do so, we define ($\mathbf{a} = n_1 \mathbf{a}_1 + n_2 \mathbf{a}_2$)

$$t(\mathbf{a}) |\mathbf{r}_{R\alpha}, \alpha\rangle = e^{i\theta_\alpha^\chi(\mathbf{a}, \mathbf{r}_{R\alpha})} |\mathbf{r}_{R\alpha} - \mathbf{a}, \alpha\rangle, \quad \langle \mathbf{r}_{R\alpha}, \alpha | t^\dagger(\mathbf{a}) = e^{-i\theta_\alpha^\chi(\mathbf{a}, \mathbf{r}_{R\alpha})} \langle \mathbf{r}_{R\alpha} - \mathbf{a}, \alpha |, \quad (13)$$

where $\theta_\alpha^\chi(\mathbf{a}, \mathbf{r}_{R\alpha})$ are the phases to be determined. Then the Hamiltonian transforms as

$$\begin{aligned} t(\mathbf{a}) H_A t^\dagger(\mathbf{a}) &= \sum_{R\alpha} \sum_{R'\alpha'} e^{-i\chi \frac{(\mathbf{r}_{R\alpha} + \mathbf{r}_{R'\alpha'})_x (\mathbf{r}_{R\alpha} - \mathbf{r}_{R'\alpha'})_y}{2l_B^2} + i\theta_\alpha^\chi(\mathbf{a}, \mathbf{r}_{R\alpha}) - i\theta_{\alpha'}^\chi(\mathbf{a}, \mathbf{r}_{R'\alpha'})} \\ &\quad \times h_{\alpha\alpha'}(\mathbf{r}_{R\alpha} - \mathbf{r}_{R'\alpha'}) |\mathbf{r}_{R\alpha} - \mathbf{a}, \alpha\rangle \langle \mathbf{r}_{R'\alpha'} - \mathbf{a}, \alpha'|. \end{aligned} \quad (14)$$

Similar to the continuum Hofstadter translation operators, we require the Hamiltonian to be commutative with magnetic translation operators on the lattice $t(\mathbf{a})$, i.e., $t(\mathbf{a})H_{\mathbf{A}}t^\dagger(\mathbf{a}) = H_{\mathbf{A}}$, so we must have

$$\begin{aligned} & \chi \frac{(\mathbf{r}_{\mathbf{R}\alpha} + \mathbf{r}_{\mathbf{R}'\alpha'})_x (\mathbf{r}_{\mathbf{R}\alpha} - \mathbf{r}_{\mathbf{R}'\alpha'})_y}{2l_B^2} - \theta_\alpha^\chi(\mathbf{a}, \mathbf{r}_{\mathbf{R}\alpha}) + \theta_{\alpha'}^\chi(\mathbf{a}, \mathbf{r}_{\mathbf{R}'\alpha'}) = \chi \frac{(\mathbf{r}_{\mathbf{R}\alpha} + \mathbf{r}_{\mathbf{R}'\alpha'} - 2\mathbf{a})_x (\mathbf{r}_{\mathbf{R}\alpha} - \mathbf{r}_{\mathbf{R}'\alpha'})_y}{2l_B^2} \\ \implies & \theta_\alpha^\chi(\mathbf{a}, \mathbf{r}_{\mathbf{R}\alpha}) - \theta_{\alpha'}^\chi(\mathbf{a}, \mathbf{r}_{\mathbf{R}'\alpha'}) = \chi \frac{(\mathbf{a})_x (\mathbf{r}_{\mathbf{R}\alpha} - \mathbf{r}_{\mathbf{R}'\alpha'})_y}{l_B^2}. \end{aligned} \quad (15)$$

So we can set

$$\theta_\alpha^\chi(\mathbf{a}, \mathbf{r}_{\mathbf{R}\alpha}) = \chi \frac{1}{l_B^2} (\mathbf{a})_x (\mathbf{r}_{\mathbf{R}\alpha})_y. \quad (16)$$

However, by acting on $|\mathbf{r}_{\mathbf{R}\alpha}, \alpha\rangle$ we can check

$$\begin{aligned} & t(\mathbf{R}_1)t(\mathbf{R}_2)|\mathbf{r}_{\mathbf{R}\alpha}, \alpha\rangle = e^{i\chi(\mathbf{R}_2)_x(\mathbf{r}_{\mathbf{R}\alpha})_y/l_B^2} t(\mathbf{R}_1)|\mathbf{r}_{\mathbf{R}\alpha} - \mathbf{R}_2, \alpha\rangle \\ & = e^{i\chi(\mathbf{R}_2)_x(\mathbf{r}_{\mathbf{R}\alpha})_y/l_B^2} e^{i\chi(\mathbf{R}_1)_x(\mathbf{r}_{\mathbf{R}\alpha} - \mathbf{R}_2)_y/l_B^2} |\mathbf{r}_{\mathbf{R}\alpha} - \mathbf{R}_1 - \mathbf{R}_2, \alpha\rangle \\ & = e^{-i\chi(\mathbf{R}_1)_x(\mathbf{R}_2)_y/l_B^2} e^{i\chi(\mathbf{R}_1 + \mathbf{R}_2)_x(\mathbf{r}_{\mathbf{R}\alpha})_y/l_B^2} |\mathbf{r}_{\mathbf{R}\alpha} - \mathbf{R}_1 - \mathbf{R}_2, \alpha\rangle \\ & = e^{-i\chi(\mathbf{R}_1)_x(\mathbf{R}_2)_y/l_B^2} t(\mathbf{R}_1 + \mathbf{R}_2)|\mathbf{r}_{\mathbf{R}\alpha}, \alpha\rangle. \end{aligned} \quad (17)$$

So $t(\mathbf{R}_1)t(\mathbf{R}_2) = e^{-i\chi(\mathbf{R}_1)_x(\mathbf{R}_2)_y/l_B^2} t(\mathbf{R}_1 + \mathbf{R}_2)$, and $t(\mathbf{R}_1)t(\mathbf{R}_2) = e^{-i\chi[(\mathbf{R}_1)_x(\mathbf{R}_2)_y - (\mathbf{R}_2)_x(\mathbf{R}_1)_y]/l_B^2} t(\mathbf{R}_2)t(\mathbf{R}_1)$, i.e., the magnetic translation operators $t(\mathbf{R}_1)$ and $t(\mathbf{R}_2)$ do not generally commute. We can also prove

$$t^n(\mathbf{a}) = \exp\left[-i\chi \frac{n(n-1)}{2} \frac{(\mathbf{a})_x(\mathbf{a})_y}{l_B^2}\right] t(n\mathbf{a}). \quad (18)$$

As a summary, we obtain the *magnetic translation operator*

$$\begin{aligned} & t(\mathbf{a})|\mathbf{r}_{\mathbf{R}\alpha}, \alpha\rangle = e^{i\chi(\mathbf{a})_x(\mathbf{r}_{\mathbf{R}\alpha})_y/l_B^2} |\mathbf{r}_{\mathbf{R}\alpha} - \mathbf{a}, \alpha\rangle, \\ & \langle\mathbf{r}_{\mathbf{R}\alpha}, \alpha|t^\dagger(\mathbf{a}) = e^{-i\chi(\mathbf{a})_x(\mathbf{r}_{\mathbf{R}\alpha})_y/l_B^2} \langle\mathbf{r}_{\mathbf{R}\alpha} - \mathbf{a}, \alpha|, \end{aligned} \quad (19)$$

which commute with the magnetic Hamiltonian

$$t(\mathbf{a})H_{\mathbf{A}}t^{-1}(\mathbf{a}) = H_{\mathbf{A}}. \quad (20)$$

However, in general $t(\mathbf{R}_1)$ and $t(\mathbf{R}_2)$ do not commute:

$$t(\mathbf{R}_1)t(\mathbf{R}_2) = e^{-i\chi(\mathbf{R}_1 \times \mathbf{R}_2)_z/l_B^2} t(\mathbf{R}_2)t(\mathbf{R}_1). \quad (21)$$

Notice also that we generally have $t^\dagger(\mathbf{R}) = t^{-1}(\mathbf{R}) \neq t(-\mathbf{R})$. Instead, we have

$$t^\dagger(\mathbf{R}) = t^{-1}(\mathbf{R}) = e^{-i\chi(\mathbf{R})_x(\mathbf{R})_y/l_B^2} t(-\mathbf{R}). \quad (22)$$

1.3 Magnetic supercell, Brillouin zone, and Bloch sum basis

For $\mathbf{R}_1 = \mathbf{a}_1$, $\mathbf{R}_2 = \mathbf{a}_2$ (from now on we suppose $(\mathbf{a}_1 \times \mathbf{a}_2)_z > 0$), Eq. (21) gives

$$t(\mathbf{a}_1)t(\mathbf{a}_2) = e^{-i2\pi\Phi/\Phi_0} t(\mathbf{a}_2)t(\mathbf{a}_1), \quad \Phi = \chi B(\mathbf{a}_1 \times \mathbf{a}_2)_z \quad (23)$$

where Φ is the (signed) magnetic flux in the lattice unit cell, and $\Phi_0 = h/e$ is the flux quantum.

If the flux ratio is rational (here p and q are coprime positive integers)

$$\frac{\Phi}{\Phi_0} = \chi \frac{p}{q}, \quad \gcd(p, q) = 1, \quad (24)$$

one can enlarge the unit cell by a factor of q : $\mathbf{L}_1 = q\mathbf{a}_1$, $\mathbf{L}_2 = \mathbf{a}_2$ with the supercell area $(\mathbf{L}_1 \times \mathbf{L}_2)_z = 2\pi l_B^2 p$ (please be careful that in continuum case we define the magnetic supercell $\mathbf{m}_1 = q\mathbf{a}_1/p$, $\mathbf{m}_2 = \mathbf{a}_2$ with

the supercell area $2\pi l_B^2$, and then the commensurate supercell $\mathbf{L}_1 = p\mathbf{m}_1$, $\mathbf{L}_2 = \mathbf{m}_2$), so that the magnetic translation operators $t(\mathbf{L}_1)$ and $t(\mathbf{L}_2)$ commute

$$t(\mathbf{L}_1)t(\mathbf{L}_2) = t(q\mathbf{a}_1)t(\mathbf{a}_2) = e^{-i\chi 2\pi p}t(\mathbf{a}_2)t(q\mathbf{a}_1) = t(\mathbf{L}_2)t(\mathbf{L}_1), \quad (25)$$

and they commute with the Hamiltonian H_A . We call $\mathbf{L}_1$ and $\mathbf{L}_2$ the unit vectors of the commensurate supercell. The corresponding magnetic reciprocal basis vectors are $\mathbf{P}_1 = \mathbf{b}_1/q$ and $\mathbf{P}_2 = \mathbf{b}_2$ where $\mathbf{b}_i$ are reciprocal basis vectors of the original lattice ($\mathbf{b}_i \cdot \mathbf{a}_j = 2\pi\delta_{ij}$). For the momentum $\mathbf{k} = k_1\mathbf{P}_1 + k_2\mathbf{P}_2 = k_1\mathbf{b}_1/q + k_2\mathbf{b}_2$ within the supercell Brillouin zone (SBZ), where $k_1 \in [0, 1)$, $k_2 \in [0, 1)$, and the “supercell sublattice index” $s = 0, 1, \dots, q-1$, we define the supercell Bloch sum basis as

$$\begin{aligned} |\mathbf{k}, s, \alpha\rangle &= \frac{1}{\sqrt{N/q}} \sum_{n_1, n_2} e^{i\mathbf{k} \cdot (n_1\mathbf{L}_1 + n_2\mathbf{L}_2)} t^{n_1}(\mathbf{L}_1) t^{n_2}(\mathbf{L}_2) |s\mathbf{a}_1 + \boldsymbol{\tau}_\alpha, \alpha\rangle \\ &= \frac{1}{\sqrt{N/q}} \sum_{n_1, n_2} e^{i\mathbf{k} \cdot (n_1\mathbf{L}_1 + n_2\mathbf{L}_2)} e^{-i\chi \frac{n_1(n_1-1)}{2} \frac{(\mathbf{L}_1)_x(\mathbf{L}_1)_y}{l_B^2} - i\chi \frac{n_2(n_2-1)}{2} \frac{(\mathbf{L}_2)_x(\mathbf{L}_2)_y}{l_B^2}} t(n_1\mathbf{L}_1) t(n_2\mathbf{L}_2) |s\mathbf{a}_1 + \boldsymbol{\tau}_\alpha, \alpha\rangle, \end{aligned} \quad (26)$$

where N is the number of original lattice unit cells, and thus N/q denotes the number of magnetic unit cells. Then one can clearly show that

$$\begin{aligned} t(\mathbf{L}_1)|\mathbf{k}, s, \alpha\rangle &= e^{-i\mathbf{k} \cdot \mathbf{L}_1} |\mathbf{k}, s, \alpha\rangle = e^{-i2\pi k_1} |\mathbf{k}, s, \alpha\rangle, \\ t(\mathbf{L}_2)|\mathbf{k}, s, \alpha\rangle &= e^{-i\mathbf{k} \cdot \mathbf{L}_2} |\mathbf{k}, s, \alpha\rangle = e^{-i2\pi k_2} |\mathbf{k}, s, \alpha\rangle. \end{aligned} \quad (27)$$

The index $s = 0, \dots, q-1$ should not be confused with the continuum LL-basis label $r = 0, \dots, p-1$. The former labels original unit cells inside the real-space supercell, while the latter labels the p copies of the commensurate SBZ inside the flux-one LL MBZ.

1.4 Hamiltonian matrix elements

The Hamiltonian matrix elements in the magnetic Bloch sum basis Eq. (26) can be calculated as

$$\begin{aligned} H_{s\alpha, s'\alpha'}(\mathbf{k}) &= \langle \mathbf{k}, s, \alpha | H_A | \mathbf{k}, s', \alpha' \rangle \\ &= \frac{1}{N/q} \sum_{n_1 n_2} \sum_{n'_1 n'_2} e^{i\mathbf{k} \cdot [(n'_1 - n_1)\mathbf{L}_1 + (n'_2 - n_2)\mathbf{L}_2]} \langle s\mathbf{a}_1 + \boldsymbol{\tau}_\alpha, \alpha | t^{-n_2}(\mathbf{L}_2) t^{-n_1}(\mathbf{L}_1) H_A t^{n'_1}(\mathbf{L}_1) t^{n'_2}(\mathbf{L}_2) | s'\mathbf{a}_1 + \boldsymbol{\tau}_{\alpha'}, \alpha' \rangle \\ &= \frac{1}{N/q} \sum_{n_1 n_2} \sum_{n'_1 n'_2} e^{i\mathbf{k} \cdot [(n'_1 - n_1)\mathbf{L}_1 + (n'_2 - n_2)\mathbf{L}_2]} \langle s\mathbf{a}_1 + \boldsymbol{\tau}_\alpha, \alpha | H_A t^{n'_1 - n_1}(\mathbf{L}_1) t^{n'_2 - n_2}(\mathbf{L}_2) | s'\mathbf{a}_1 + \boldsymbol{\tau}_{\alpha'}, \alpha' \rangle \\ &= \sum_{n_1 n_2} e^{i\mathbf{k} \cdot (n_1\mathbf{L}_1 + n_2\mathbf{L}_2)} \langle s\mathbf{a}_1 + \boldsymbol{\tau}_\alpha, \alpha | H_A t^{n_1}(\mathbf{L}_1) t^{n_2}(\mathbf{L}_2) | s'\mathbf{a}_1 + \boldsymbol{\tau}_{\alpha'}, \alpha' \rangle \\ &= \sum_{n_1 n_2} e^{i\mathbf{k} \cdot (n_1\mathbf{L}_1 + n_2\mathbf{L}_2)} e^{-i\chi \frac{n_1(n_1-1)}{2} \frac{(\mathbf{L}_1)_x(\mathbf{L}_1)_y}{l_B^2} - i\chi \frac{n_2(n_2-1)}{2} \frac{(\mathbf{L}_2)_x(\mathbf{L}_2)_y}{l_B^2}} \langle s\mathbf{a}_1 + \boldsymbol{\tau}_\alpha, \alpha | H_A t(n_1\mathbf{L}_1) t(n_2\mathbf{L}_2) | s'\mathbf{a}_1 + \boldsymbol{\tau}_{\alpha'}, \alpha' \rangle \\ &= \sum_{n_1 n_2} e^{i\mathbf{k} \cdot (n_1\mathbf{L}_1 + n_2\mathbf{L}_2)} e^{-i\chi \frac{n_1(n_1-1)}{2} \frac{(\mathbf{L}_1)_x(\mathbf{L}_1)_y}{l_B^2} - i\chi \frac{n_2(n_2-1)}{2} \frac{(\mathbf{L}_2)_x(\mathbf{L}_2)_y}{l_B^2}} e^{i\chi n_1 \frac{(\mathbf{L}_1)_x(s'\mathbf{a}_1 - n_2\mathbf{L}_2 + \boldsymbol{\tau}_{\alpha'})_y}{l_B^2} + i\chi n_2 \frac{(\mathbf{L}_2)_x(s'\mathbf{a}_1 + \boldsymbol{\tau}_{\alpha'})_y}{l_B^2}} \\ &\quad \times \langle s\mathbf{a}_1 + \boldsymbol{\tau}_\alpha, \alpha | H_A | s'\mathbf{a}_1 - n_1\mathbf{L}_1 - n_2\mathbf{L}_2 + \boldsymbol{\tau}_{\alpha'}, \alpha' \rangle \\ &= \sum_{n_1 n_2} e^{i\mathbf{k} \cdot (n_1\mathbf{L}_1 + n_2\mathbf{L}_2)} e^{-i\chi \frac{n_1(n_1-1)}{2} \frac{(\mathbf{L}_1)_x(\mathbf{L}_1)_y}{l_B^2} - i\chi \frac{n_2(n_2-1)}{2} \frac{(\mathbf{L}_2)_x(\mathbf{L}_2)_y}{l_B^2}} e^{i\chi n_1 \frac{(\mathbf{L}_1)_x(s'\mathbf{a}_1 - n_2\mathbf{L}_2 + \boldsymbol{\tau}_{\alpha'})_y}{l_B^2} + i\chi n_2 \frac{(\mathbf{L}_2)_x(s'\mathbf{a}_1 + \boldsymbol{\tau}_{\alpha'})_y}{l_B^2}} \\ &\quad \times e^{-i\chi \frac{[(s+s')\mathbf{a}_1 - n_1\mathbf{L}_1 - n_2\mathbf{L}_2 + \boldsymbol{\tau}_\alpha + \boldsymbol{\tau}_{\alpha'}]_x [(s-s')\mathbf{a}_1 + n_1\mathbf{L}_1 + n_2\mathbf{L}_2 + \boldsymbol{\tau}_\alpha - \boldsymbol{\tau}_{\alpha'}]_y}{2l_B^2}} h_{\alpha\alpha'}((s-s')\mathbf{a}_1 + n_1\mathbf{L}_1 + n_2\mathbf{L}_2 + \boldsymbol{\tau}_\alpha - \boldsymbol{\tau}_{\alpha'}), \end{aligned} \quad (28)$$

where Eq. (18) has been used. This is the general Hamiltonian matrix that can be directly used numerically. Although Eq. (28) looks complicated, the summation usually survives for only several n_1, n_2 due to the cutoff

of the hopping. We notice that, if $a_{2x}a_{1y} = 0$, then $(\mathbf{a}_1 \times \mathbf{a}_2)_z = a_{1x}a_{2y} = 2\pi l_B^2 p/q$, we have

$$\begin{aligned} & H_{s\alpha, s'\alpha'}(\mathbf{k}) \\ &= \sum_{n_1 n_2} e^{i2\pi(n_1 k_1 + n_2 k_2)} e^{-i\chi\pi p q n_1(n_1-1)\frac{a_{1y}}{a_{2y}} - i\chi\pi\frac{p}{q}n_2(n_2-1)\frac{a_{2x}}{a_{1x}}} e^{i\chi 2\pi p n_1 \left[s'\frac{a_{1y}}{a_{2y}} - n_2 + \frac{(\tau_{\alpha'})_y}{a_{2y}} \right] + i\chi 2\pi\frac{p}{q}n_2\frac{a_{2x}}{a_{1x}} \left[s'\frac{a_{1y}}{a_{2y}} + \frac{(\tau_{\alpha'})_y}{a_{2y}} \right]} \\ & \times e^{-i\chi\pi\frac{p}{q} \left[s+s'-n_1q-n_2\frac{a_{2x}}{a_{1x}} + \frac{(\tau_{\alpha}+\tau_{\alpha'})_x}{a_{1x}} \right] \left[(s-s'+n_1q)\frac{a_{1y}}{a_{2y}} + n_2 + \frac{(\tau_{\alpha}-\tau_{\alpha'})_y}{a_{2y}} \right]} h_{\alpha\alpha'}((s-s'+n_1q)\mathbf{a}_1 + n_2\mathbf{a}_2 + \boldsymbol{\tau}_{\alpha} - \boldsymbol{\tau}_{\alpha'}). \end{aligned} \quad (29)$$

To simplify the algebra further, one may rotate the lattice so that

$$\mathbf{a}_1 = (a_{1x}, a_{1y}), \quad \mathbf{a}_2 = (0, a_{2y}) \parallel \mathbf{e}_y, \quad (30)$$

where $a_{1x} > 0$. Then $(\mathbf{a}_1 \times \mathbf{a}_2)_z = a_{1x}a_{2y} = 2\pi l_B^2 p/q$. Then the Bloch sum basis is

$$|\mathbf{k}, s, \alpha\rangle = \frac{1}{\sqrt{N/q}} \sum_{n_1 n_2} e^{i\mathbf{k} \cdot (n_1 q \mathbf{a}_1 + n_2 \mathbf{a}_2)} e^{-i\chi\pi p q n_1(n_1-1)\frac{a_{1y}}{a_{2y}}} t(n_1 q \mathbf{a}_1) t(n_2 \mathbf{a}_2) |s\mathbf{a}_1 + \boldsymbol{\tau}_{\alpha}, \alpha\rangle, \quad (31)$$

and the Hamiltonian matrix elements are $(\mathbf{k} = k_1 \mathbf{P}_1 + k_2 \mathbf{P}_2 = k_1 \mathbf{b}_1/q + k_2 \mathbf{b}_2)$

$$\begin{aligned} H_{s\alpha, s'\alpha'}(\mathbf{k}) &= \sum_{n_1 n_2} e^{i2\pi(n_1 k_1 + n_2 k_2)} e^{-i\chi\pi p q n_1(n_1-1)\frac{a_{1y}}{a_{2y}} + i\chi 2\pi p n_1 \left[s'\frac{a_{1y}}{a_{2y}} - n_2 + \frac{(\tau_{\alpha'})_y}{a_{2y}} \right]} \\ & \times e^{-i\chi\pi\frac{p}{q} \left[s+s'-n_1q + \frac{(\tau_{\alpha}+\tau_{\alpha'})_x}{a_{1x}} \right] \left[(s-s'+n_1q)\frac{a_{1y}}{a_{2y}} + n_2 + \frac{(\tau_{\alpha}-\tau_{\alpha'})_y}{a_{2y}} \right]} h_{\alpha\alpha'}((s-s'+n_1q)\mathbf{a}_1 + n_2\mathbf{a}_2 + \boldsymbol{\tau}_{\alpha} - \boldsymbol{\tau}_{\alpha'}). \end{aligned} \quad (32)$$

Or, one may rotate the lattice so that

$$\mathbf{a}_1 = (a_{1x}, 0) \parallel \mathbf{e}_x, \quad \mathbf{a}_2 = (a_{2x}, a_{2y}), \quad (33)$$

where $a_{2y} > 0$. Then $(\mathbf{a}_1 \times \mathbf{a}_2)_z = a_{1x}a_{2y} = 2\pi l_B^2 p/q$. Then the Bloch sum basis is

$$|\mathbf{k}, s, \alpha\rangle = \frac{1}{\sqrt{N/q}} \sum_{n_1 n_2} e^{i\mathbf{k} \cdot (n_1 q \mathbf{a}_1 + n_2 \mathbf{a}_2)} e^{-i\chi\pi\frac{p}{q}n_2(n_2-1)\frac{a_{2x}}{a_{1x}}} t(n_1 q \mathbf{a}_1) t(n_2 \mathbf{a}_2) |s\mathbf{a}_1 + \boldsymbol{\tau}_{\alpha}, \alpha\rangle, \quad (34)$$

and the Hamiltonian matrix elements are $(\mathbf{k} = k_1 \mathbf{P}_1 + k_2 \mathbf{P}_2 = k_1 \mathbf{b}_1/q + k_2 \mathbf{b}_2)$

$$\begin{aligned} H_{s\alpha, s'\alpha'}(\mathbf{k}) &= \sum_{n_1 n_2} e^{i2\pi(n_1 k_1 + n_2 k_2)} e^{-i\chi\pi\frac{p}{q}n_2(n_2-1)\frac{a_{2x}}{a_{1x}} + i\chi 2\pi p n_1 \left[-n_2 + \frac{(\tau_{\alpha'})_y}{a_{2y}} \right] + i\chi 2\pi\frac{p}{q}n_2\frac{a_{2x}}{a_{1x}} \frac{(\tau_{\alpha'})_y}{a_{2y}}} \\ & \times e^{-i\chi\pi\frac{p}{q} \left[s+s'-n_1q-n_2\frac{a_{2x}}{a_{1x}} + \frac{(\tau_{\alpha}+\tau_{\alpha'})_x}{a_{1x}} \right] \left[n_2 + \frac{(\tau_{\alpha}-\tau_{\alpha'})_y}{a_{2y}} \right]} h_{\alpha\alpha'}((s-s'+n_1q)\mathbf{a}_1 + n_2\mathbf{a}_2 + \boldsymbol{\tau}_{\alpha} - \boldsymbol{\tau}_{\alpha'}). \end{aligned} \quad (35)$$

2 Examples

2.1 Square lattice

For square lattice with $\mathbf{a}_1 = a\mathbf{e}_x$, $\mathbf{a}_2 = a\mathbf{e}_y$, the magnetic translation operators are (using $a^2/l_B^2 = 2\pi p/q$)

$$\begin{aligned} t(\mathbf{L}_1)|x_1\mathbf{a}_1 + x_2\mathbf{a}_2\rangle &= e^{i\chi q a^2 x_2/l_B^2} |(x_1 - q)\mathbf{a}_1 + x_2\mathbf{a}_2\rangle = T(\mathbf{L}_1)|x_1\mathbf{a}_1 + x_2\mathbf{a}_2\rangle, \\ t(\mathbf{L}_2)|x_1\mathbf{a}_1 + x_2\mathbf{a}_2\rangle &= |x_1\mathbf{a}_1 + (x_2 - 1)\mathbf{a}_2\rangle = T(\mathbf{L}_2)|x_1\mathbf{a}_1 + x_2\mathbf{a}_2\rangle. \end{aligned} \quad (36)$$

We see the magnetic translation operators $t(\mathbf{L}_1)$ and $t(\mathbf{L}_2)$ reduce to the original lattice translation operators $T(\mathbf{L}_1)$ and $T(\mathbf{L}_2)$, respectively. This in general only happens for the square lattice (for example, for graphene in the next example, in general $t(\mathbf{L}_1) \neq T(\mathbf{L}_1)$ or $t(\mathbf{L}_2) \neq T(\mathbf{L}_2)$).

The Hamiltonian matrix elements are

$$H_{ss'}(\mathbf{k}) = \sum_{n_1 n_2} e^{i2\pi(n_1 k_1 + n_2 k_2)} e^{-i\chi\pi\frac{p}{q}n_2(s+s'+n_1q)} h((s-s'+n_1q)\mathbf{a}_1 + n_2\mathbf{a}_2). \quad (37)$$

Up to nearest neighbors (NN), only $h(\pm \mathbf{a}_1) = h(\pm \mathbf{a}_2) = t$ survives, so we get

$$\begin{aligned} H_{ss'}(\mathbf{k}) &= t \sum_{n_1} e^{i2\pi n_1 k_1} (\delta_{s-s'+n_1 q, 1} + \delta_{s-s'+n_1 q, -1}) + \delta_{ss'} t \sum_{n_2} e^{i2\pi n_2 (k_2 - \chi s \frac{p}{q})} (\delta_{n_2, 1} + \delta_{n_2, -1}) \\ &= t \sum_{n_1} e^{i2\pi n_1 k_1} (\delta_{s-s'+n_1 q, 1} + \delta_{s-s'+n_1 q, -1}) + 2\delta_{ss'} t \cos \left[2\pi \left(k_2 - \chi s p / q \right) \right]. \end{aligned} \quad (38)$$

The first term represents the hopping along x (supercell) direction, and the second term represents the hopping along y direction.

2.2 Massive Dirac cone: staggered honeycomb lattice

We next consider the honeycomb lattice with a staggered sublattice potential. We choose the primitive vectors in the second convention above,

$$\mathbf{a}_1 = a(1, 0), \quad \mathbf{a}_2 = a \left(\frac{1}{2}, \frac{\sqrt{3}}{2} \right), \quad (39)$$

so that $\mathbf{a}_1 \parallel \mathbf{e}_x$ and $(\mathbf{a}_1 \times \mathbf{a}_2)_z = \frac{\sqrt{3}}{2} a^2 = 2\pi l_B^2 \frac{p}{q}$. Here a is the length of the triangular Bravais primitive vector. The honeycomb NN bond length is therefore $a/\sqrt{3}$. We take two sublattices A, B in each unit cell,

$$\boldsymbol{\tau}_A = (0, 0), \quad \boldsymbol{\tau}_B = \left(0, \frac{a}{\sqrt{3}} \right). \quad (40)$$

The staggered sublattice potential is $h_{AA}(0) = -h_{BB}(0) = M$. The nearest-neighbor hopping is

$$h_{AB}(\boldsymbol{\delta}_i) = -t, \quad h_{BA}(-\boldsymbol{\delta}_i) = -t \quad (41)$$

for $\boldsymbol{\delta}_1 = \boldsymbol{\tau}_A - \boldsymbol{\tau}_B = \left(0, -\frac{a}{\sqrt{3}} \right)$, $\boldsymbol{\delta}_2 = \mathbf{a}_2 + \boldsymbol{\tau}_A - \boldsymbol{\tau}_B = \left(\frac{a}{2}, \frac{a}{2\sqrt{3}} \right)$, and $\boldsymbol{\delta}_3 = -\mathbf{a}_1 + \mathbf{a}_2 + \boldsymbol{\tau}_A - \boldsymbol{\tau}_B = \left(-\frac{a}{2}, \frac{a}{2\sqrt{3}} \right)$, all with length $a/\sqrt{3}$. Using the general formula above, the magnetic Bloch Hamiltonian has dimension $2q \times 2q$, with matrix indices (s, α) and (s', α') , where $s, s' = 0, \dots, q-1$, $\alpha, \alpha' = A, B$. For a hopping labeled by $\mathbf{d} = \ell_1 \mathbf{a}_1 + \ell_2 \mathbf{a}_2 + \boldsymbol{\tau}_\alpha - \boldsymbol{\tau}_{\alpha'} = (s - s' + n_1 q) \mathbf{a}_1 + n_2 \mathbf{a}_2 + \boldsymbol{\tau}_\alpha - \boldsymbol{\tau}_{\alpha'}$, the corresponding magnetic-cell integers are $s = s' + \ell_1 \bmod q$, $n_1 = (\ell_1 + s' - s)/q$, $n_2 = \ell_2$. Using $\frac{a_{2x}}{a_{1x}} = \frac{1}{2}$, the Hamiltonian Eq. (35) is

$$\begin{aligned} H_{s\alpha, s'\alpha'}(\mathbf{k}) &= \sum_{n_1, n_2} e^{i2\pi(n_1 k_1 + n_2 k_2)} e^{-i\chi\pi\frac{p}{q}\frac{n_2(n_2-1)}{2}} e^{i\chi 2\pi p n_1 \left[-n_2 + \frac{(\boldsymbol{\tau}_{\alpha'})_y}{a_{2y}} \right] + i\chi\pi\frac{p}{q} n_2 \frac{(\boldsymbol{\tau}_{\alpha'})_y}{a_{2y}}} \\ &\times e^{-i\chi\pi\frac{p}{q} \left[s + s' - n_1 q - \frac{n_2}{2} + \frac{(\boldsymbol{\tau}_\alpha + \boldsymbol{\tau}_{\alpha'})_x}{a} \right] \left[n_2 + \frac{(\boldsymbol{\tau}_\alpha - \boldsymbol{\tau}_{\alpha'})_y}{a_{2y}} \right]} h_{\alpha\alpha'}((s - s' + n_1 q) \mathbf{a}_1 + n_2 \mathbf{a}_2 + \boldsymbol{\tau}_\alpha - \boldsymbol{\tau}_{\alpha'}). \end{aligned} \quad (42)$$

For the present staggered honeycomb model, the only nonzero terms are the two onsite mass terms and the six nearest-neighbor hoppings listed above.